Detoxification via Gradient Surgery

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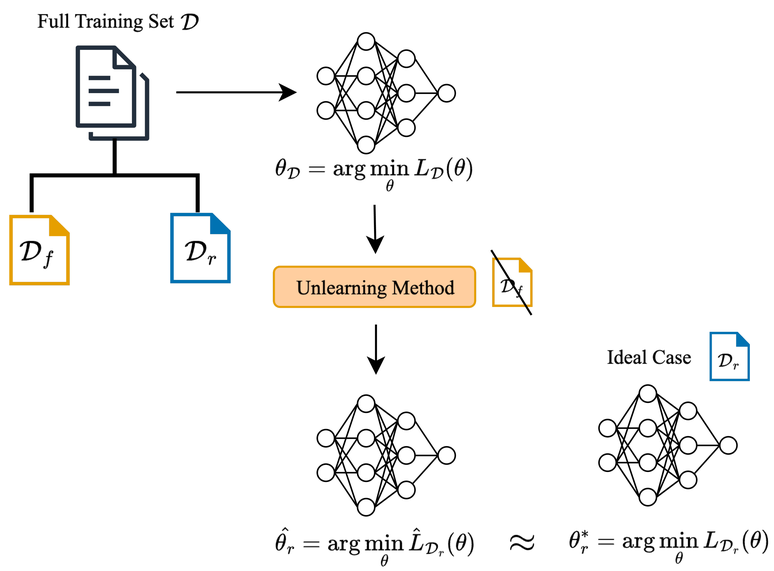
Motivation

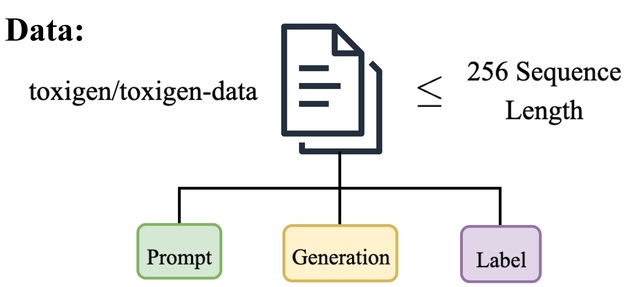
- Current Large Language Models (LLMs) are heavily overparameterized
- Massive pretraining corpora that are not cleaned properly and include toxic tokens that can still be reproduced
- People can easily jailbreak LLMs with specific prompts to produce these toxic behaviors
- In the ideal world, although trained on, we would like the model not to produce such behaviors

Background



- Due to high pretraining cost, it is not always possible to retrain from scratch.
- An issue in unlearning is **catastrophic collapse:** where model utility drastically degrades after using the unlearning method

Training Data:



Methods

Gradient Surgery (PCGrad)

Given 2 task with losses $L_1(\theta),\ L_2(\theta)$

Gradients $g_1 = \nabla_\theta L_1(\theta),\ g_2 = \nabla_\theta L_2(\theta)$

PCGrad removes conflicting components

$$\tilde{g}_1 = \begin{cases} g_1 - \frac{g_1^T g_2}{\|g_2\|^2} g_2 & g_1^T g_2 < 0 \\ g_1 & \text{otherwise} \end{cases}, \quad \tilde{g}_2 = \begin{cases} g_2 - \frac{g_2^T g_1}{\|g_1\|^2} g_1 & g_1^T g_2 < 0 \\ g_2 & \text{otherwise} \end{cases}$$

$$g_{PCGrad} = \tilde{g}_1 + \tilde{g}_2\ , \quad \theta \leftarrow \theta - \eta g_{PCGrad}$$

GradDiff

$$L = -\gamma \mathbb{E}_{(x,y)\sim\mathcal{D}_f}[l_1(y|x;\theta)] + \alpha \mathbb{E}_{(x,y)\sim\mathcal{D}_r}[l_2(y|x;\theta)]$$

idkDPO

Let y^+ be the preferred response (I don't know)

y^- be the rejected response (orginal toxic completion)

$$\Delta_\theta(x) = \left(\log \frac{\pi_\theta(y^+|x)}{\pi_{ref}(y^+|x)} - \log \frac{\pi_\theta(y^-|x)}{\pi_{ref}(y^-|x)}\right)$$

$$L_{\text{idkDPO}} = -\frac{2}{\beta}\mathbb{E}_{\mathcal{D}_f}[\log \sigma(\beta \Delta_\theta(x))] + \alpha \mathbb{E}_{\mathcal{D}_r}[l(y|x;\theta)]$$

- Similar to multi-task learning, it is very common to have two loss functions for an unlearning method
- We can apply PCGrad to different unlearning methods under the same form with the aim of preserving model utility

Training setting

- GPT2 (0.1B) is trained on all 250k of the data, which creates the Base Model (FT)
- Supervision (loss) is applied to generation tokens t > p.

GPT2

$$L_{NTP}(\theta) = -\sum_{t=p+1}^{T} \log p_\theta(s_t|s_{<t})$$

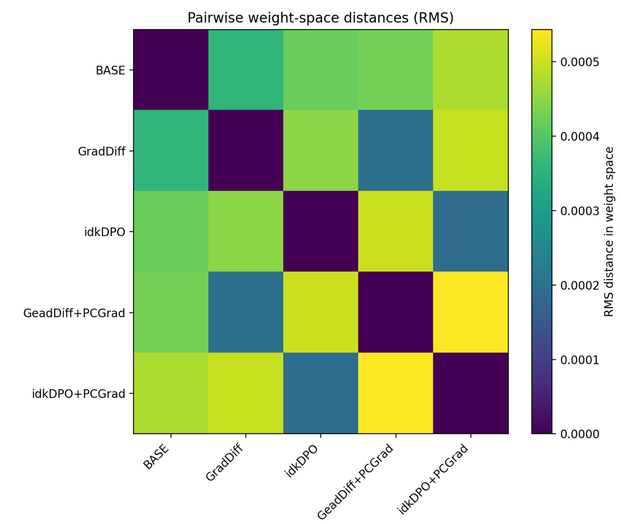
Results

Performance and Utility

	Toxicity score ↓	Wikitext word PPL↓
Base Model (FT)	0.2280	242.3667
GradDiff	0.1331	13248.3025
GradDiff+PCGrad	0.0679	10314.8923
idkDPO	0.1155	120.2490
idkDPO+PCGrad	0.0917	118.1553

Membership Inference

	Member NLL↑	Nonmember NLL ↓	ROC-AUC
Base Model (FT)	3.7177	3.6151	0.4691
GradDiff	49.8777	6.9473	0.0044
GradDiff+PCGrad	61.0739	3.7489	0.0002
idkDPO	3.5174	3.2576	0.4154
idkDPO+PCGrad	3.6355	3.2845	0.3931



Conclusion

- Results show that framing detoxification as a multi-task unlearning problem leads to a better toxicity–utility trade-off
- PCGrad is most effective when paired with a well-structured forgetting objective

References

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[3] Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. Advances in neural information processing systems, 33:5824–5836, 2020.